

Surname	Centre Number	Candidate Number
Other Names		0


GCSE – NEW
3445U10-1


APPLIED SCIENCE (Double Award)
Unit 1: Energy, Resources and the Environment
FOUNDATION TIER

WEDNESDAY, 14 JUNE 2017 – MORNING
1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	3	
2.	8	
3.	13	
4.	14	
5.	18	
6.	19	
Total	75	

 3445U101
01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator, pencil and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

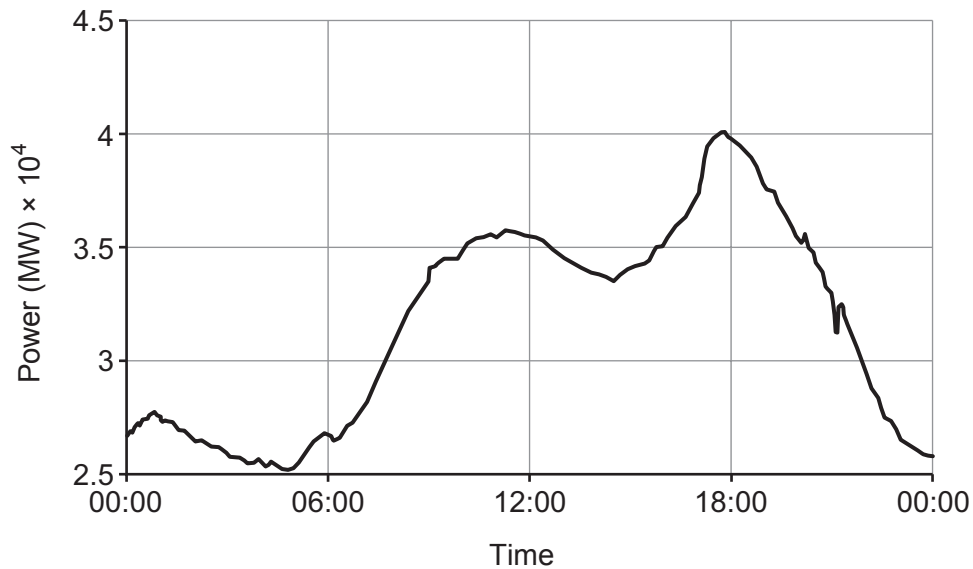
Question 5(a)(ii) is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A periodic table is printed on page 20.

Answer **all** the questions in the spaces provided.

1. The graph below shows how demand for electricity from the National Grid changed over a 24 h period on 23rd October 2011.



- (a) Describe how demand changed between 12:00 and 18:00.

[2]

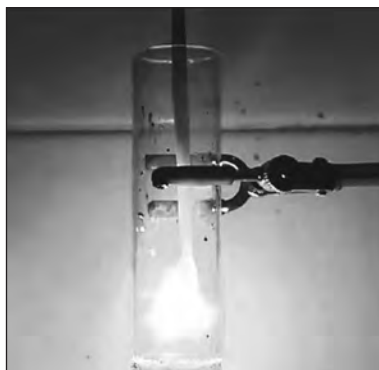
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- (b) Hydroelectric power stations only produce electricity when there is a surge in demand. However water needs to be pumped back up at some time during the 24 h period when demand is below 3×10^4 MW and falling. Mark an **X** on the graph to show where this could be done.

[1]

2. Magnesium oxide helps prevent scale formation in boilers, heat exchangers, and piping. Burning magnesium in a gas jar of oxygen produces magnesium oxide on a small scale.



- (a) The symbol for oxygen is $^{16}_8\text{O}$.

Complete the following statements about an atom of oxygen.

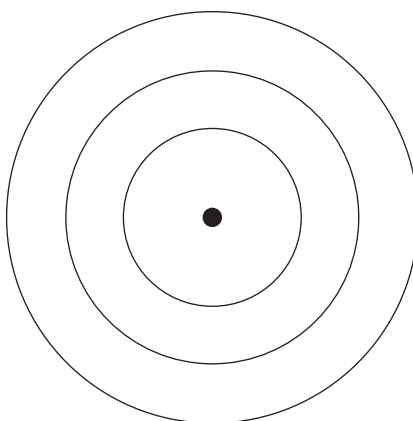
[3]

An oxygen atom contains electrons.

An oxygen atom contains neutrons.

An oxygen atom contains protons.

- (b) An atom of magnesium contains 12 electrons. Complete the diagram to show the electronic structure of a magnesium atom. [1]



- (c) State what happens to a magnesium atom when it becomes a magnesium ion.

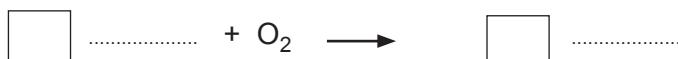
[1]

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- (d) **Complete** the balanced symbol equation for the following reaction.

[3]

magnesium + oxygen $\longrightarrow$ magnesium oxide



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3. The halogens are a group of elements found in group 7 of the periodic table. The table below gives some information about the elements in this group.

Halogen	Atomic number	Melting point (°C)	Boiling point (°C)	Radius of the atom (units)
fluorine	9	-219	-188	(0.64) / 1.1 / 1.7
chlorine	17	(-5 / -101 / -220)	-34	0.99
bromine	35	-7	59	(1.14 / 1.45 / 1.56)
iodine	(22 / 31 / 53)	114	184	1.33
astatine	85	302	380	1.40

- (a) **Complete** the table above by **circling** the correct value in each of the brackets. One has been done as an example. [3]

- (b) Use the information in the table and your own knowledge to answer the following questions.

- (i) The boiling points of the alkali metals (group 1) decreases from the top to the bottom of their group. Describe whether the halogens follow the same trend. [2]

.....

.....

.....

- (ii) State the **two** halogens in the table which are gases at 20°C. [2]

..... and

- (iii) State which halogen is a liquid at 20°C. [1]

.....

- (iv) State what happens to the size of the halogen atoms as you go down from the top to the bottom of the group. [1]

.....

(v) **Tick (✓)** the box next to the correct statements about halogens.

[3]

Fluorine has the highest melting point

☐

The most reactive halogen is fluorine

☐

Astatine has the most electrons

☐

Iodine is a solid at 20°C

☐

The most reactive halogen is astatine

☐

Chlorine boils at a lower temperature than fluorine

☐

(c) Sodium chloride can be made by burning sodium in chlorine gas. However it is found naturally in rock salt, which can be obtained from the Earth's crust. **Tick (✓)** the box next to the correct process for extraction of rock salt from the Earth's crust.

[1]

fracking

☐

deep shaft mining

☐

fractional distillation

☐

drilling

☐

4. Rising gas prices are affecting many households. The government can help households keep their gas bills as low as possible.

Data for a typical house are shown below. The table also shows government estimates for the current usage of each method.

Part of house	Heat loss before insulation is fitted (%)	Method of reducing heat loss	Cost of method (£)	Annual saving on gas bills (£)	% of homes already using this method
roof	25	loft insulation	450	150	91
windows & doors	10	double glazing	3 900	90	70
walls	40	cavity wall insulation	1 200	250	42
floor	15	carpets	800	40	90
draughts	10	draught excluders	60	5	85

- (a) A heating engineer estimates keeping the house with no insulation at 20°C results in gas bills of £2 000 per year.

Use the information above to answer the following questions.

- (i) Calculate the cost of the heat lost **through the roof**. [2]

cost = £

- (ii) State how the engineer's estimate will change if the home is kept at 22°C. [1]

.....

- (iii) State how the engineer's estimate will change if the price of gas drops. [1]

.....

- (iv) The engineer recommends that loft insulation should be installed first.

Calculate the payback time for loft insulation. [2]

payback time = years

- (v) Which method of reducing heat loss should the government help people install by giving them a grant? Give **one** reason for your choice. [2]

.....

.....

.....

- (b) At present, many homes with central heating lack a timer, sufficient hot water tank insulation, room thermostat and thermostatic radiator valves. Upgrading all homes so they have these four features could make savings of over 25% of the energy used for heating and hot water.

Fitting new hot water tank insulation, a room thermostat and thermostatic radiator valves reduces the annual gas bill for a house from £2 000 to £1 400.

- (i) Calculate the percentage saving made to the gas bill. [2]

% saving =

- (ii) Use the equation:

$$\text{decrease in the number of units used} = \frac{\text{savings}}{\text{cost of one unit}}$$

to calculate the decrease in the number of units used. [2]
Assume each unit costs £0.20.

decrease in the number of units used = kWh

- (c) State **two** ways in which using less gas to heat the house will benefit the environment. [2]

1.

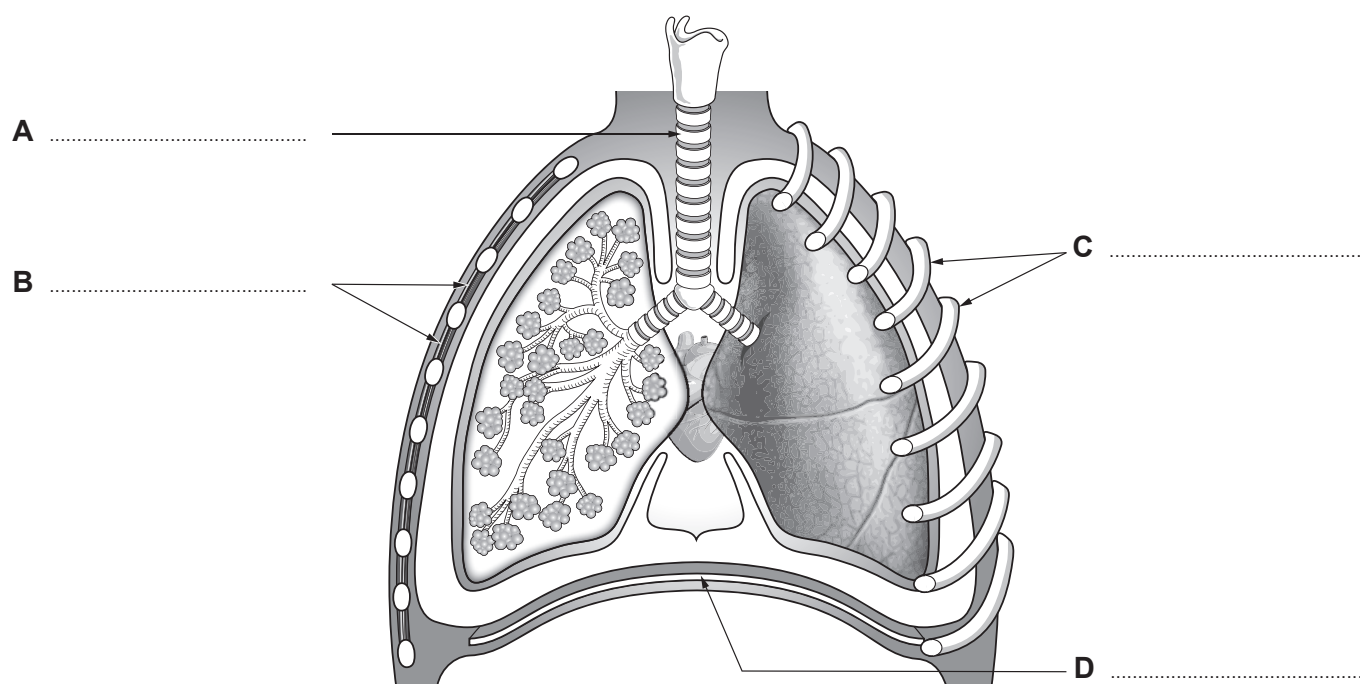
2.

5. Sports science students are learning about the respiratory system.

(a) (i) Label the diagram below using words from the box.

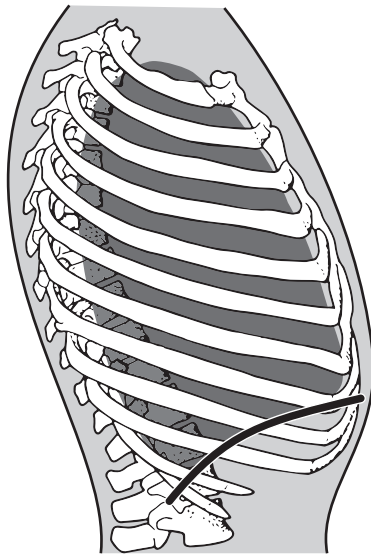
[4]

ribs intercostal muscles diaphragm lung trachea

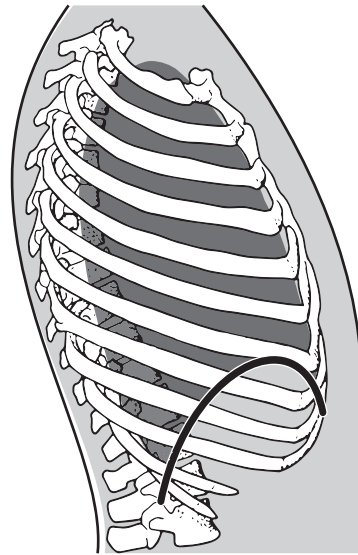


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[6 QER]



breathing in

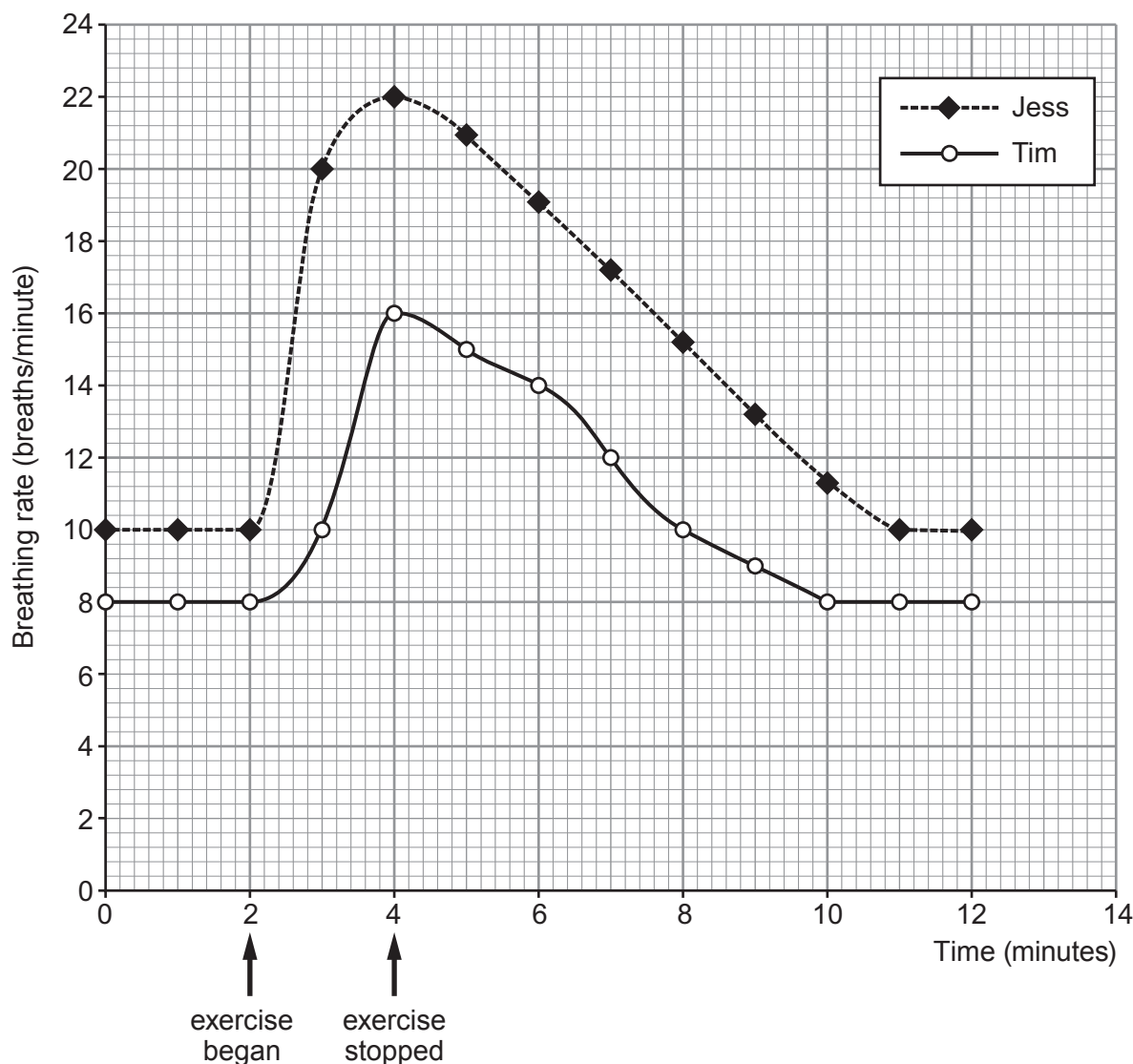


breathing out

- (b) Two students, Tim and Jess, studied the effect of exercise on breathing rate. They followed the method below:

1. record breathing rate before exercising
2. exercise for 2 minutes by running on a treadmill
3. measure the breathing rates during and after exercise up until 12 minutes

Their results are shown on the graph below.



Use the graph to complete the following sentences.

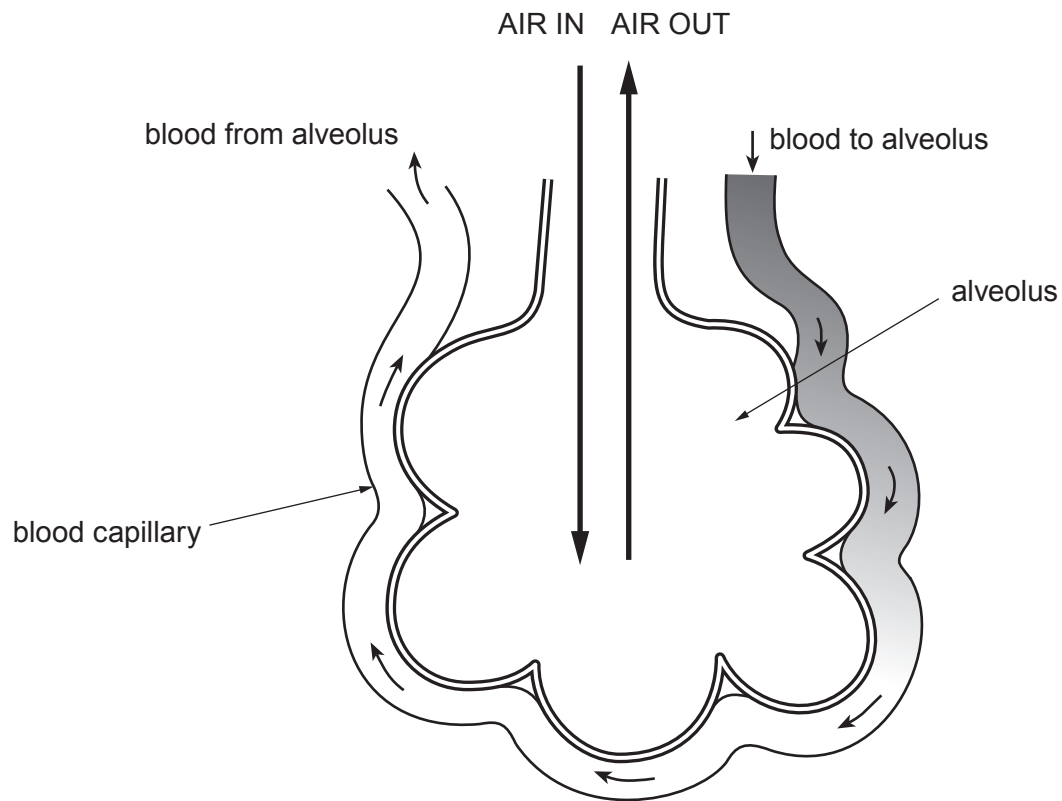
[3]

At 2 minutes, the difference between Tim and Jess' breathing rate is breaths per minute.

Between 8 and 10 minutes Tim's breathing rate drops than Jess' breathing rate.

Tim's breathing rate returns to normal at minutes.

- (c) The diagram below is of an alveolus within the lung where gases are exchanged.



- (i) State which gases are exchanged.

[2]

..... and

- (ii) Name the process by which these gases move between the alveolus and the capillary.

[1]

.....

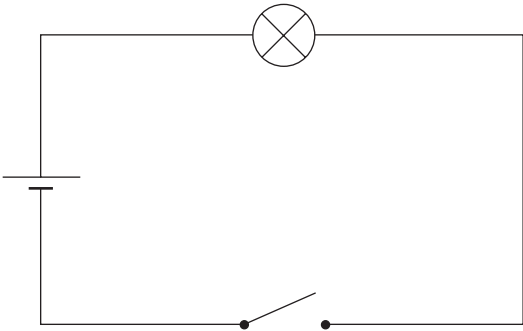
- (iii) State **two** ways in which the alveoli are adapted for gas exchange.

[2]

1.

2.

6. Students are investigating how the current through a lamp varies with the voltage across it.



- (a) (i) Part of the circuit used is shown above. **Add** a variable resistor, ammeter and voltmeter to the circuit diagram. [3]
- (ii) Describe the method you would use to take a range of results. [3]

.....

.....

.....

.....

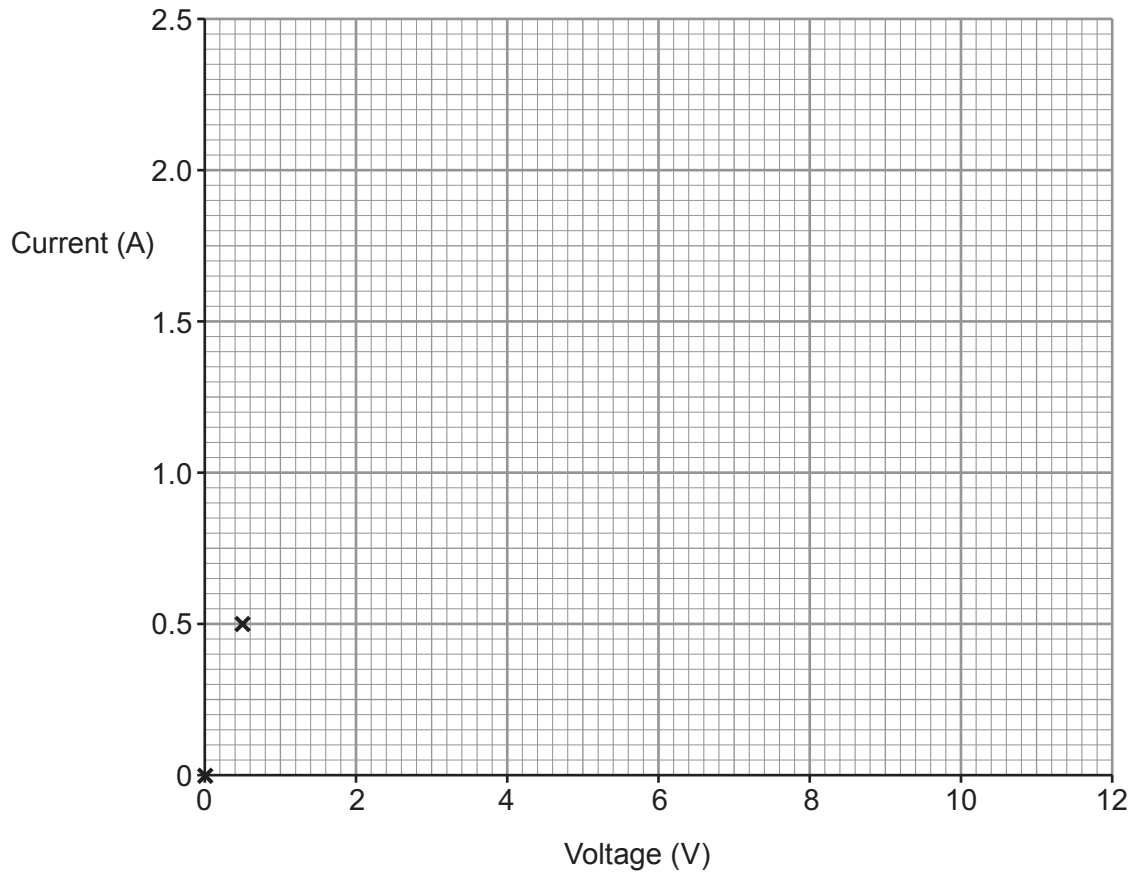
(b) Results from the experiment are shown below.

Voltage (V)	Current (A)
0.0	0.0
0.5	0.5
2.0	0.8
4.0	1.1
6.0	1.4
8.0	1.6
10.0	1.8
12.0	1.9

- (i) Plot the data on the grid below and draw a suitable line.

[3]

The first two points have been plotted for you.



- (ii) Describe the relationship between current and voltage for the lamp.

[2]

.....

.....

.....

- (c) (i) Use the equation:

$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the lamp at 0.5 V.

[2]

resistance = Ω

- (ii) State how the resistance of the lamp varies between 0.5 V and 12 V.

[1]

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- (d) (i) Use the equation:

$$\text{power} = \text{voltage} \times \text{current}$$

to calculate the power of the lamp at 0.5 V.

[2]

power = W

- (ii) State how the power of the lamp varies between 0.5 V and 12 V.

[1]

- (e) The experiment was repeated with a length of wire at constant temperature. The current through the wire was 1.5 A when the voltage across it was 12 V.

Add a line to your graph to show how current varies with voltage for this wire.

[2]

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THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1																								4 He Helium 2											
		9 Be Beryllium 4										11 B Boron 5		12 C Carbon 6		14 N Nitrogen 7		16 O Oxygen 8		19 F Fluorine 9		20 Ne Neon 10													
23 Na Sodium 11		24 Mg Magnesium 12												27 Al Aluminium 13		28 Si Silicon 14		31 P Phosphorus 15		32 S Sulfur 16		35.5 Cl Chlorine 17		40 Ar Argon 18											
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21		48 Ti Titanium 22		51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30		70 Ga Gallium 31		73 Ge Germanium 32		75 As Arsenic 33		79 Se Selenium 34		80 Br Bromine 35		84 Kr Krypton 36	
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39		91 Zr Zirconium 40		93 Nb Niobium 41		96 Mo Molybdenum 42		99 Tc Technetium 43		101 Ru Ruthenium 44		103 Rh Rhodium 45		106 Pd Palladium 46		108 Ag Silver 47		112 Cd Cadmium 48		115 In Indium 49		119 Sn Tin 50		122 Sb Antimony 51		128 Te Tellurium 52		127 I Iodine 53		131 Xe Xenon 54	
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57		179 Hf Hafnium 72		181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80		204 Tl Thallium 81		207 Pb Lead 82		209 Bi Bismuth 83		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86	
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																															

Key

1
H
Hydrogen
1

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number